Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 270 (2025) 1788–1795

Procedia Computer Science

www.elsevier.com/locate/procedia

29th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2025)

Data-Driven Transparency: Machine Learning and Social Network Analysis for Corruption Detection in Public Procurement

Ricardo Muñoz-Cancino[a], Sebastián A. Ríos[b,*]

[a]X-Company SpA., Santiago 8170178, Chile

[b]Business Intelligence Research Center (CEINE), Industrial Engineering Department, University of Chile, Beauchef 851, Santiago 8370456, Chile

Abstract

Corruption in public procurement is a global challenge that can cause inefficiencies, financial losses, and a decline in public trust. Fraudulent behaviors, such as supplier collusion or conflicts of interest, undermine the integrity, transparency, and integrity of procurement processes. Traditional monitoring mechanisms struggle to detect these situations due to the complexity and volume of procurement processes. This paper proposes a methodology to detect corrupt behavior in public procurement using Machine Learning and Social Network Analysis. Our model is trained to identify suspicious patterns in supplier relationships, improving due diligence and risk assessment. As a case study, we apply our methodology to Chilean public procurement data, demonstrating its potential to improve transparency and mitigate corruption risks.

© 2025 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the KES International.

Keywords: Public Procurement; Machine Learning ; Anomaly Detection; Social Network Analysis

1. Introduction

Public procurement represents a significant proportion of government spending and is exposed to risks such as corruption, collusion, favoritism, and conflicts of interest. According to the OECD [10], approximately 57% of the bribery cases are related to government procurement, highlighting the need for effective mechanisms to detect and prevent irregularities. In Chile, public procurement represents around 6% of GDP, with a transaction volume reaching $15 billion through ChileCompra in 2023 [2]. ChileCompra is the official platform of the Chilean government for public procurement. Its primary role is to facilitate and make public sector procurement processes transparent, ensuring that government institutions acquire goods and services competitively and comply with legal regulations [2].

Despite several reforms [1], monitoring corruption in these processes is heavily based on manual audits and traditional mechanisms, limiting the response to detect potential irregularities. Corruption in public procurement refers to

* Corresponding author.
E-mail address: srios@dii.uchile.cl

1877-0509 © 2025 The Authors. Published by Elsevier B.V.
This is an open access article under the CC BY-NC-ND license (https://creativecommons.org/licenses/by-nc-nd/4.0)
Peer-review under responsibility of the scientific committee of the KES International.
10.1016/j.procs.2025.09.299

illegal or unethical practices in the government's procurement of goods and services. These practices include bribery, collusion between suppliers, bid rigging, favoritism, and fraud in contract execution. Corruption in this area generates cost overruns, affects the quality of goods and services, undermines trust in government institutions, and interferes with economic and social development. Although various initiatives are being taken to improve transparency in public procurement [4], there is still a gap in analyzing the connections between the different actors involved in these processes. Detecting suspicious behavior requires evaluating each supplier in isolation and understanding their links with other actors within the public procurement ecosystem. However, traditional detection methodologies struggle to identify complex patterns of collusion or conflicts of interest. This study proposes a framework based on machine learning and social network analysis to detect corruption-related behaviors in public procurement. Our model aims to identify anomalous patterns and connections between suppliers that may suggest irregular practices. Using historical data from ChileCompra, we explore how this methodology can improve public spending supervision, providing a more efficient and data-driven approach to detecting corruption risks. The structure of this paper is as follows. Section 2 reviews the relevant literature and background information. Section 3 describes the dataset used in this study. Section 4 presents the proposed methodology. Section 5 presents the findings and discusses their implications. Finally, Section 6 summarizes the key findings and proposes future research efforts.

2. Background and Related Work

Public procurement has traditionally been an undervalued study area despite its importance in government management and economic development [17]. Efforts to improve procurement processes' transparency and efficiency have applied various strategies and theoretical models for decades. In particular, benchmarking tools have been proposed to evaluate and optimize the quality of procurement processes [11]. Furthermore, Snider and Rendon developed conceptual models to understand and design more effective procurement policies [16]. Fraud in public procurement can be explained through theories such as the diamond fraud model [5], which considers factors such as pressure, opportunity, rationalization, and the individual capacity to commit fraud. Rustiarini et al. has used this conceptual framework to understand the incentives and conditions that facilitate corruption in government procurement [14]. In recent years, artificial intelligence techniques have gained relevance in detecting fraud in public procurement. The application of machine learning algorithms, neural networks, natural language processing, and network analysis has improved the identification of fraudulent patterns in public procurement processes [9]. Lyra et al. present a systematic review of data-driven methods for detecting fraud and collusion in government procurement, highlighting network science and risk analysis's usefulness in investigating these crimes [7]. Additionally, decision support systems have enabled the automation of corruption risk analysis in procurement. A system developed in Brazil combines data mining, graph theory, and cluster analysis to identify fraud patterns in public procurement, facilitating the work of law enforcement agencies [18]. Another approach has been using machine learning algorithms to detect collusion in public auctions to identify anticompetitive behavior [13].

3. Data Description

The data used in this study were downloaded from the ChileCompra website, which provides a parametric link to obtain information on bidding processes by month: `https://transparenciachc.blob.core.windows.net/lic-da/{year}-{month}.zip`. For this analysis, data for the entire year 2023 were collected. The dataset includes 2,317,034 records, corresponding to 122,570 tenders. Each record represents a bid submitted by a supplier in a given tender and includes information about the successful supplier in each case. From these data, a network was constructed where each node represents a supplier, and two suppliers are connected if they participated in the same tender. The resulting network consists of 42,282 nodes and 836,426 connections (edges), with 792 connected components and an average degree of 39.5. The largest connected component is composed of 39,581 nodes and 829,664 connections.

4. Proposed methodology

Our methodology is based on the KDD process [3], combining machine learning and social network analysis (SNA) techniques. We have applied this approach in community detection [12], influencer identification [15] and credit risk

analysis [8], and in this case, it is adapted to detect anomalous behavior in public procurement. The process consists of the following stages:

1. **Data Gathering:** The data was obtained from the ChileCompra platform, where suppliers' records of tenders and bids submitted during 2023 were downloaded. Each record represents a supplier's bid in a given tender.

2. **Data Transformation and Network Configuration:** From the collected data, a network is constructed in which each node represents a supplier, and connections (edges) are established between suppliers participating in the same tender. This way, competitive relationships and potential collusion patterns between suppliers are modeled.

3. **Network and Node Feature Extraction:** The network structure and supplier behavior within the public procurement ecosystem are characterized by various metrics. These metrics are extracted at the connected component and node levels. In this paper, we extract:
   (a) **Connected component features**
      - **Number of Nodes:** Number of suppliers in the connected component.
      - **Number of Edges:** Total number of connections within the connected component.
      - **Average Degree:** Average degree of the nodes within the connected component.
      - **Density:** Proportion of existing connections relative to the total possible connections.
      - **Diameter:** Maximum distance (in number of links) between two nodes within the connected component.
   (b) **Node (supplier) features in the connected component**
      - **Degree:** Number of direct connections of the node (number of bids shared with other suppliers).
      - **Clustering Coefficient:** A node's level of clustering within the network indicates the probability that its neighbors are connected.
      - **Betweenness Centrality:** A measure of betweenness that indicates how many shortest paths pass through the node.
      - **Eigenvector Centrality:** A node's relevance based on the importance of its connections.
      - **PageRank:** An adaptation of Google's algorithm that measures a node's influence in the network.
      - **Hub Score:** Indicates whether a node tends to connect with influential suppliers.
      - **Authority Score:** Measures how frequently other influential suppliers reference a node.

4. **Build the Final Dataset:** Two distinct datasets are generated; the first is a connected component-level dataset that contains aggregated metrics for each component. The second one in a node (supplier)-level dataset, constructed by combining the characteristics of each supplier in the network with relevant financial information, including:
   - Company age
   - Sales segment.
   - Number of employees.
   - Total number of tenders for which the company has applied.
   - Total amount awarded in tenders.
   - Number of public agencies with which the company has worked.

5. **Modeling:** In the modeling step to identify anomalies in the public procurement network, two Isolation Forest models are trained [6], each focusing on a different level of analysis within the network structure:
   - **Supplier Model (Node Level):** This model analyzes suppliers individually, identifying those with unusual characteristics based on their financial attributes and network metrics. Suppliers with atypical behavior, such as extreme values in centrality measures or unusual financial patterns, are detected and flagged as potential anomalies.
   - **Connected Components Model (Network Level):** This model evaluates the connected component structures within the network, identifying those with unusual characteristics in terms of the number of suppliers, connections, density, and connectivity patterns. Components with atypical structures, such as highly dense clusters or unusually large or small clusters, are classified as anomalies.

For both models, a contamination level of 5% is set, meaning that approximately 5% of observations are considered expected anomalies. Furthermore, the data is split into 90% for training and 10% for testing, allowing for evaluation of the model's performance on unseen data. The results obtained allow for the identification of both individual suppliers and network structures that could be associated with irregular behavior in public tenders.

6. **Validation and Interpretation:** The results obtained from the test dataset are analyzed, assessing the interpretability of the alerts generated by the model. The reasons for identifying certain suppliers as unusual are investigated by examining their network characteristics and financial attributes. This allows for qualitative validation of the patterns detected and a better understanding of the factors associated with potentially irregular behavior in public procurement.

5. Results and Discussion

The Isolation Forest-based anomaly detection models were successfully trained and evaluated at supplier and connected component levels. The results allow for identifying relevant patterns in the network structure and supplier characteristics that may be associated with unusual behavior in public procurement processes. Below are the results for the two models, the Supplier Model and the Connected Components Model. For each, a summary plot is presented with the results of the shap values and a description of how the most important variables influence the classification of being unusual.

5.1. Supplier Model (Nodes)

The anomaly detection model at the supplier level highlights the importance of both network structural and financial factors. Figure 1 presents the summary plot of the Supplier Model. This graph shows the model variables on the Y-axis, ordered from highest to lowest importance according to their impact on the prediction. On the X-axis, the SHAP values are shown, representing each variable's contribution to the model's prediction. To the right of each variable is a scatter plot, where each point represents an observation in the test dataset. The color of the points indicates the variable's original value, with red indicating high values and blue indicating low values.

The effect and contribution of the most important variables on the model's prediction, classifying each observation as unusual or not, is detailed below. These effects are complemented by the SHAP dependence plots presented in Figure 2. A dependence plot is a scatter plot that shows how the values of an input variable impact the model's prediction through its SHAP values. The X-axis shows the original values of the analyzed variable, and the Y-axis shows the SHAP values, which indicate the impact of that variable on the prediction. Additionally, the points are colored according to another variable in the dataset, which allows for identifying possible interactions.

- **Company Age (10.7%):** Nonlinear behavior is observed in company age, with both very new and older companies exhibiting more negative shape values, which increases their likelihood of being classified as unusual. This effect is more significant in older companies. Due to their longer history of operation, older companies may be more involved in irregular practices or have a higher likelihood of collusion due to their accumulated experience and established market relationships. On the other hand, new companies may exhibit greater instability or atypical behavior when adapting to the market.

- **Clustering Coefficient (10.5%):** Figure 2c shows the dependence plot for the Clustering Coefficient feature. The Clustering Coefficient exhibits nonlinear behavior, with low values primarily contributing to being classified as unusual. Although values close to zero also exhibit negative values, their magnitude is smaller. In this case, a low clustering coefficient means a supplier is connected to several other suppliers. However, these suppliers do not know each other, indicating a more dispersed and less cohesive structure. On the other hand, a high clustering coefficient indicates that a supplier is connected to several suppliers who also know each other and are, in turn, connected to each other. This may suggest a more closed network and possibly more prone to irregular behavior or collusion.

- **Sales Bracket (8.1%):** Figure 2a shows the dependence plot for the Sales Bracket feature. Companies with low sales levels, as do those with very high sales, tend to be more unusual. This behavior may be related to

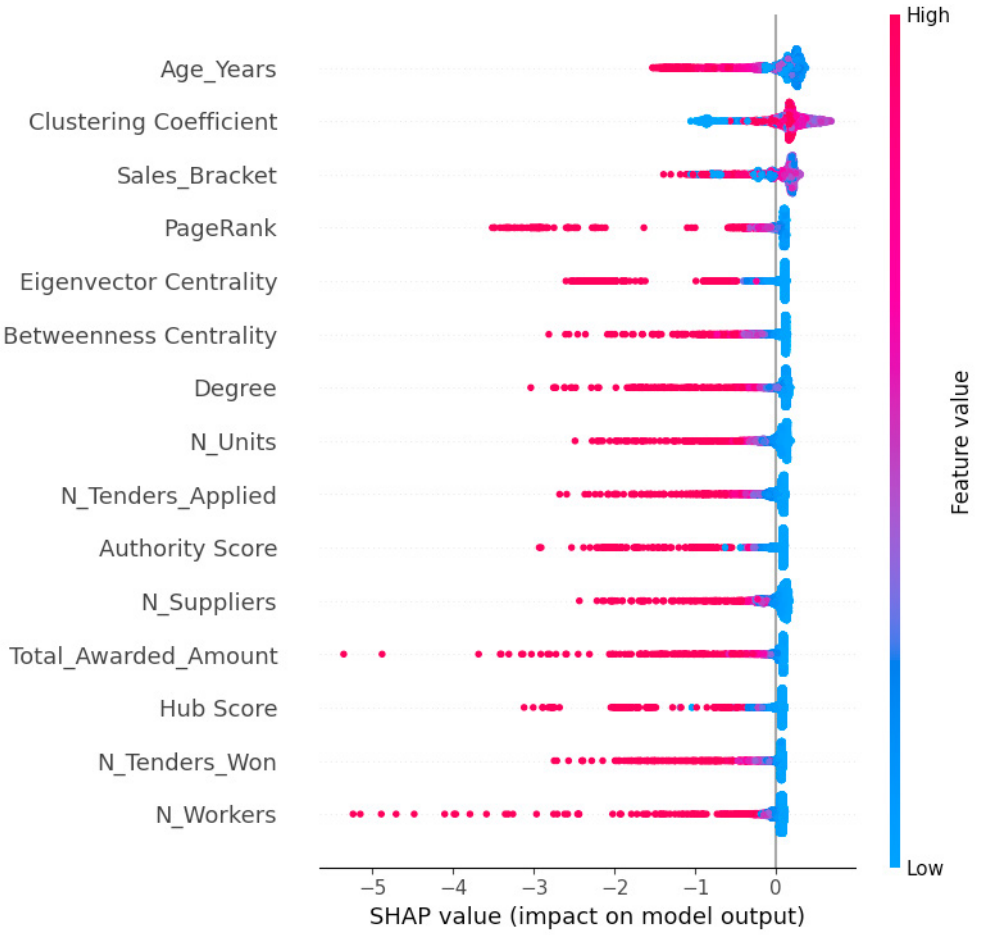


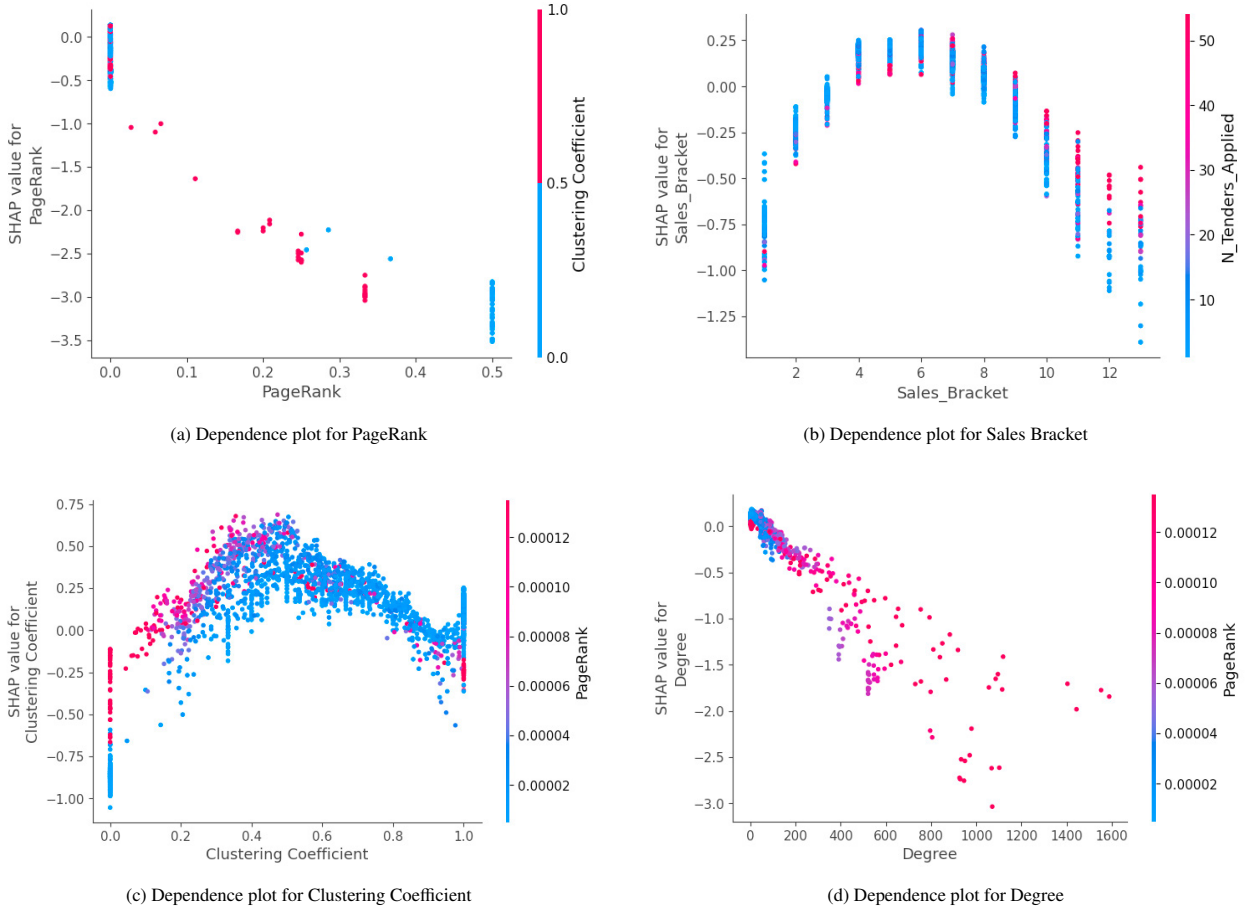
Fig. 1: Supplier Model Summary Plot: SHAP values analysis for the node level model, illustrating the impact of different features in detecting anomalous suppliers.

small companies with fewer resources or a greater disposition to commit to non-transparent practices due to competition or the need for survival. On the other hand, companies with very high sales may be involved in unethical practices due to their economic power and market influence, making them points of focus in anomaly detection models.

- **PageRank (6.9%):** Figure 2b shows the dependence plot for the PageRank feature. Suppliers with a high PageRank, that is, those who are centrally located in the network and well connected, can be classified as unusual. This behavior may be related to suppliers who, being central within the network, have more significant influence over bidding processes and may be more likely to be involved in irregular practices due to their visibility or strategic importance within the public procurement ecosystem.

- **Eigenvector Centrality (6.7%) and Betweenness Centrality (6.6%)**: Both Eigenvector Centrality and Betweenness Centrality indicate the structural influence of suppliers within the network. These variables exhibit similar behaviors to PageRank. Eigenvector Centrality measures the node importance by considering its direct connections and the connections of the nodes to which it is linked, indicating the supplier's importance within a broader network. Betweenness Centrality measures the Degree to which a node acts as an intermediary between other nodes, suggesting greater control over information or processes. Theses indicators are important for detecting key suppliers that could be involved in anomalous patterns within the public procurement network.

- **Degree (6.4%):** Figure 2d shows the dependency plot for the Degree feature. Suppliers with a high degree have lower SHAP values. Therefore, a highly connected supplier is more likely to be classified by the model as unusual.



(a) Dependence plot for PageRank

(b) Dependence plot for Sales Bracket

(c) Dependence plot for Clustering Coefficient

(d) Dependence plot for Degree

Fig. 2: SHAP dependence plots for key features in the supplier model.

These results suggest that the structural interaction of suppliers in the bidding network, combined with financial information, can identify actors with unusual patterns, which could indicate potential risks of collusion or fraud.

5.2. Connected Components Model

At the connected components level, the model highlights the relevance of metrics that describe the structure and connectivity of the network:

- **MaxEdgeAttribute (21.4%):** Connected components with higher values for edge attributes are more likely to be classified as unusual. This suggests that stronger or more significant connections between nodes within a component may indicate anomalous or unusual patterns.

- **Number of Nodes (13.6%) and Number of Edges (15.2%):** Connected components with a larger number of nodes (suppliers) or edges (connections between suppliers) are more likely to be classified as unusual. This indicates that larger components (regarding the number of nodes or connections) tend to exhibit more complex or less common behaviors, which could be related to irregular practices or non-competitive network behavior.

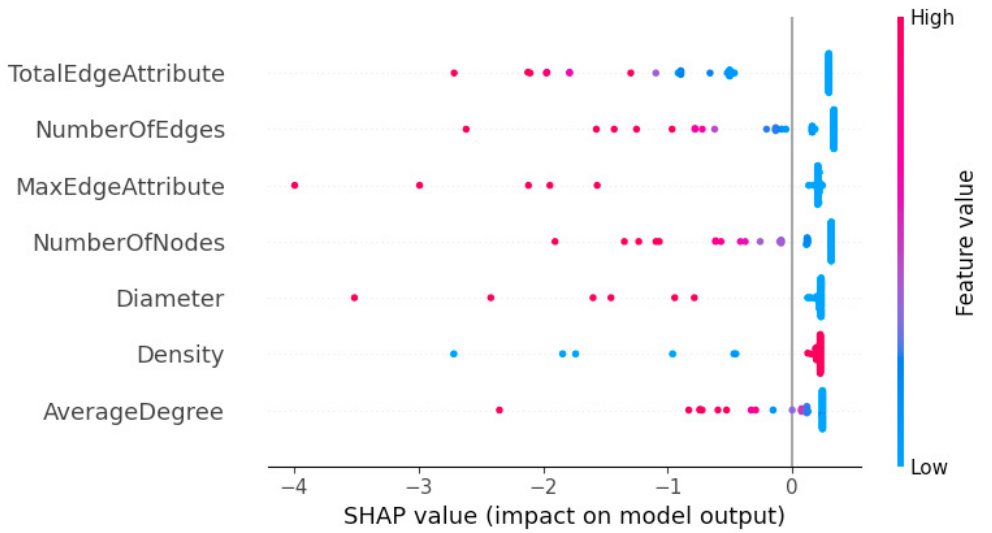


Fig. 3: Connected Components Model Summary Plot: SHAP values analysis for connected components level model, highlighting feature importance in anomaly detection.

- **Diameter (13.4%) and Density (11.8%):**The diameter of a network reflects its dispersion, that is, how ”spread out” the network is in terms of the maximum distance between two nodes. Connected components with a larger diameter tend to be classified as unusual, as greater dispersion may reflect a more fragmented network or one with less predictable connections. On the other hand, Density indicates the concentration of connections in the network. A high density suggests a more concentrated and cohesive network where many nodes are strongly connected to each other. In both diameter and density cases, higher values indicate a greater likelihood that the connected component will be classified as unusual, as they reflect more complex or anomalous structures that could indicate irregular practices.
- **Average Degree (10.7%):** The average Degree of a network reflects how many connections (edges) each node (supplier) has, on average, within the connected component. Suppliers with a higher average degree (i.e., those with multiple connections within a component) have an important role in the network structure. These providers may be influential actors within the network; therefore, a network with a higher average degree is more likely to be classified as unusual.

These findings show that supplier connectivity is a determining factor in detecting anomalies in the public procurement network, suggesting that specific interaction patterns may indicate irregular practices.

The results allow us to identify suppliers and network structures with characteristics associated with anomalous behavior. Combining Machine Learning with Social Network Analysis has proven effective in detecting non-trivial patterns in the public procurement ecosystem. The findings of this study can serve as input to improve oversight and control mechanisms in government contracting, allowing authorities to focus efforts on suppliers or networks with suspicious characteristics. Furthermore, future research could complement this approach with additional information, such as disciplinary histories or previous complaints, to refine anomaly detection capabilities.

6. Conclusions and Future Work

This study has demonstrated the effectiveness of using Machine Learning and Social Network Analysis together for anomaly detection in public procurement. Using anomaly detection models such as Isolation Forest, we have identified suppliers and network structures with unusual behavior patterns, enabling the generation of early warnings about potential corruption or collusion in government tenders. This approach prevents irregular practices in public procurement processes, improving transparency and fairness.

The results show the relevance of various network structural metrics, such as supplier centrality, the density of connections within related components, and supplier financial attributes. These metrics are important for detecting anomalous patterns that might otherwise go unnoticed. These analyses provide information for agencies responsible for control and oversight, optimizing decision-making in public procurement management, and facilitating the fight against corruption.

Although the results are promising, several opportunities exist to improve and expand this study. One key area would be integrating additional data, such as supplier background information, sanctions, or previous complaints, which could enhance the model. Furthermore, temporal network analysis is important in detecting changes and trends in contracting patterns over time. This would identify emerging dynamics in the supplier network, helping to anticipate potential irregular behavior before it materializes. Finally, validating the results with domain experts, such as government authorities and oversight agencies, is important to interpret and adjust the model's findings according to specific risk criteria. This collaboration could strengthen the system's effectiveness and ensure that the alerts generated are valuable and relevant to corruption prevention efforts.

References

[1] ChileCompra, 2024. Tender for Consultancy on Automation of Monitoring Rules. ChileCompra, Mercado Público Portal. URL: https://www.mercadopublico.cl/Procurement/Modules/RFB/DetailsAcquisition.aspx?qs=Hkz58cW8mlxiom78qtI4IQ==. tender ID 869591-7-LP24.

[2] Dirección ChileCompra, 2024. Cuenta Pública ChileCompra 2023-2024. URL: https://www.chilecompra.cl/wp-content/uploads/2024/05/Documento-CuentaPublicaChileCompra-2023-2024.pdf. Accessed: 2025-03-30.

[3] Fayyad, U., Piatetsky-Shapiro, G., Smyth, P., 1996. The kdd process for extracting useful knowledge from volumes of data. Communications of the ACM 39, 27–34.

[4] Gobierno de Chile, 2024. Estrategia Nacional de Integridad Pública. URL: https://www.integridadytransparencia.gob.cl/estrategia-nacional-de-integridad-publica/. Accessed: 2025-03-30.

[5] Khamainy, A.H., Ali, M., Setiawan, M.A., 2022. Detecting financial statement fraud through new fraud diamond model: the case of indonesia. Journal of Financial Crime 29, 925–941.

[6] Liu, F.T., Ting, K.M., Zhou, Z.H., 2012. Isolation-based anomaly detection. ACM Transactions on Knowledge Discovery from Data (TKDD) 6, 1–39.

[7] Lyra, M.S., Damásio, B., Pinheiro, F.L., Bacao, F., 2022. Fraud, corruption, and collusion in public procurement activities, a systematic literature review on data-driven methods. Applied Network Science 7, 83.

[8] Muñoz-Cancino, R., Bravo, C., Ríos, S.A., Graña, M., 2023. On the combination of graph data for assessing thin-file borrowers' creditworthiness. Expert Systems with Applications 213, 118809. URL: https://www.sciencedirect.com/science/article/pii/S0957417422018279, doi:https://doi.org/10.1016/j.eswa.2022.118809.

[9] Nai, R., Sulis, E., Meo, R., et al., 2022. Public procurement fraud detection and artificial intelligence techniques: a literature review, in: Companion Proceedings of the 23rd International Conference on Knowledge Engineering and Knowledge Management, CEUR-WS. pp. 1–13.

[10] OECD, 2025. Integrity in Public Procurement. URL: https://www.oecd.org/en/topics/sub-issues/integrity-in-public-procurement.html. accessed: 2025-03-30.

[11] Raymond, J., 2008. Benchmarking in public procurement. Benchmarking: An International Journal 15, 782–793.

[12] Ríos, S.A., Muñoz, R., 2012. Dark web portal overlapping community detection based on topic models, in: Proceedings of the ACM SIGKDD Workshop on Intelligence and Security Informatics, Association for Computing Machinery, New York, NY, USA. URL: https://doi.org/10.1145/2331791.2331793, doi:10.1145/2331791.2331793.

[13] Rodríguez, M.J.G., Rodríguez-Montequín, V., Ballesteros-Pérez, P., Love, P.E., Signor, R., 2022. Collusion detection in public procurement auctions with machine learning algorithms. Automation in Construction 133, 104047.

[14] Rustiarini, N.W., Nurkholis, N., Andayani, W., 2019. Why people commit public procurement fraud? the fraud diamond view. Journal of public procurement 19, 345–362.

[15] Ríos, S.A., Aguilera, F., Nuñez-Gonzalez, J.D., Graña, M., 2019. Semantically enhanced network analysis for influencer identification in online social networks. Neurocomputing 326-327, 71–81. URL: https://www.sciencedirect.com/science/article/pii/S0925231217315242, doi:https://doi.org/10.1016/j.neucom.2017.01.123.

[16] Snider, K.F., Rendon, R.G., 2008. Public procurement policy: Implications for theory and practice. Journal of public procurement 8, 310–333.

[17] Thai, K.V., 2001. Public procurement re-examined. Journal of public procurement 1, 9–50.

[18] Velasco, R.B., Carpanese, I., Interian, R., Paulo Neto, O.C., Ribeiro, C.C., 2021. A decision support system for fraud detection in public procurement. International Transactions in Operational Research 28, 27–47.